

Q1 - 26 July - Shift 2

The value of $\log_e 2 \frac{d}{dx} (\log_{\cos x} \operatorname{cosec} x)$ at $x = \frac{\pi}{4}$ is

Space for your notes:

- (A) $-2\sqrt{2}$ (B) $2\sqrt{2}$ (C) -4 (D) 4

Q2 - 27 July - Shift 2

For the curve $C : (x^2 + y^2 - 3) + (x^2 - y^2 - 1)^5 = 0$,
the value of $3y' - y^3 y''$, at the point (α, α) , $\alpha > 0$,
on C, is equal to _____.

Space for your notes:

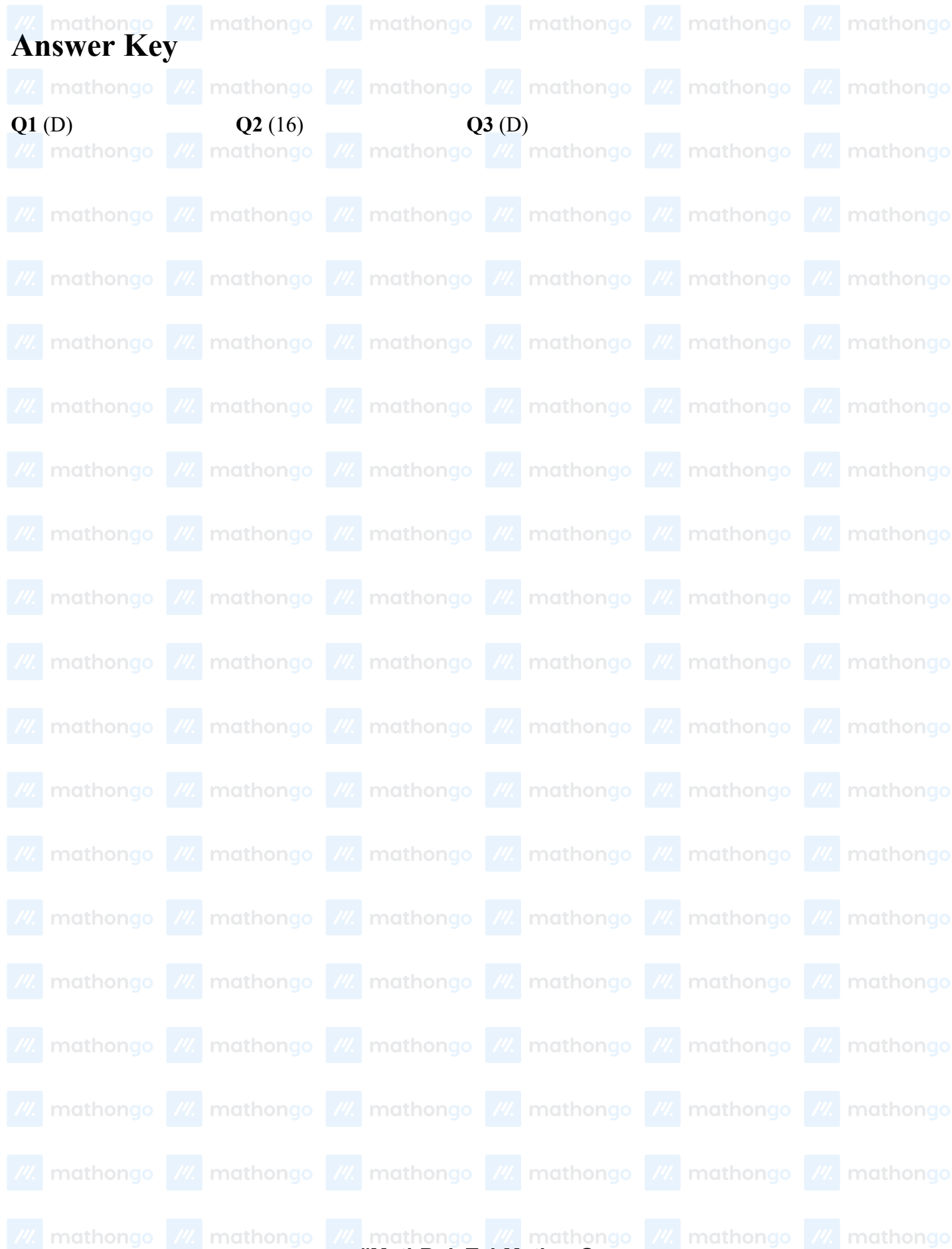
Q3 - 28 July - Shift 2

Let $x(t) = 2\sqrt{2} \cos t \sqrt{\sin 2t}$ and
 $y(t) = 2\sqrt{2} \sin t \sqrt{\sin 2t}$, $t \in \left(0, \frac{\pi}{2}\right)$. Then

Space for your notes:

$\frac{1 + \left(\frac{dy}{dx}\right)^2}{\frac{d^2 y}{dx^2}}$ at $t = \frac{\pi}{4}$ is equal to

- (A) $\frac{-2\sqrt{2}}{3}$ (B) $\frac{2}{3}$
(C) $\frac{1}{3}$ (D) $\frac{-2}{3}$

Answer Key**Q1 (D)****Q2 (16)****Q3 (D)****#MathBoleTohMathonGo**

Q1 (D)

$$\log_e 2 \frac{d}{dx} (\log_{\cos x} \operatorname{cosec} x)$$

Let,

$$y = \log_{\cos x} \operatorname{cosec} x$$

$$y = -\frac{\ln(\sin x)}{\ln(\cos x)}$$

$$\frac{dy}{dx} = -\frac{[\cot x \cdot \ln(\cos x) + \tan x \cdot \ln(\sin x)]}{(\ln(\cos x))^2}$$

$$\left. \frac{dy}{dx} \right|_{x=\frac{\pi}{4}} = \frac{4}{\ln 2}$$

Now,

$$\Rightarrow \log_e 2 \cdot \frac{4}{\ln 2} = 4$$

Q2 (16)

(α, α) lies on

$$C: x^2 + y^2 - 3 + x^2 - y^2 - 1^5 = 0$$

$$\text{Put } (\alpha, \alpha), 2\alpha^2 - 3 + -1^5 = 0$$

$$\Rightarrow \alpha = \sqrt{2}$$

Now, differentiate C

$$2x + 2y \cdot y' + 5(x^2 - y^2 - 1)^4 (2x - 2yy') = 0 \dots (1)$$

$$\text{At } (\sqrt{2}, \sqrt{2})$$

$$\sqrt{2} + \sqrt{2}y' + 5(-1)^4 (\sqrt{2} - \sqrt{2}y') = 0$$

$$\Rightarrow y' = \frac{3}{2} \dots (2)$$

Diff. (1) w.r.t. x

Again, Diff. (1) w.r.t. x

$$1 + (y')^2 + yy'' + 20(x^2 - y^2 - 1)^3 (x - yy')^2 \cdot 2$$

$$+ 5(x^2 - y^2 - 1)^4 (1 - (y')^2 - yy'') = 0$$

$$\text{At } (\sqrt{2}, \sqrt{2}) \text{ and } y' = \frac{3}{2}$$

We have,

$$\left(1 + \frac{9}{4}\right) + \sqrt{2}y'' - 40\left(\sqrt{2} - \sqrt{2} \cdot \frac{3}{2}\right)^2$$

$$+ 5(1)\left(1 - \frac{9}{4} - \sqrt{2}y''\right) = 0$$

$$\Rightarrow 4\sqrt{2}y'' = -23$$

$$\therefore 3y' - y^3 y'' = \frac{9}{2} + \frac{23}{2} = 16$$

Q3 (D)

$$x = 2\sqrt{2} \cos t \sqrt{\sin 2t}$$

$$\frac{dx}{dt} = \frac{2\sqrt{2} \cos 3t}{\sqrt{\sin 2t}}$$

$$y(t) = 2\sqrt{2} \sin t \sqrt{\sin 2t}$$

$$\frac{dy}{dt} = \frac{2\sqrt{2} \sin 3t}{\sqrt{\sin 2t}}$$

$$\frac{dy}{dx} = \tan 3t$$

$$\frac{dy}{dx} = -1 \text{ at } t = \frac{\pi}{4}$$

$$\frac{d^2y}{dx^2} = \frac{3}{2\sqrt{2}} \sec^3 3t \cdot \sqrt{\sin 2t} = -3 \text{ at } t = \frac{\pi}{4}$$

$$\therefore \frac{1 + \left(\frac{dy}{dx}\right)^2}{\frac{d^2y}{dx^2}} = \frac{1+1}{-3} = -\frac{2}{3}$$